Exercise #1

Consider the 3R planar robot in the configuration $\boldsymbol{q}$ shown in Fig. 1, controlled by the joint velocity $\dot{\boldsymbol{q}} \in \mathbb{R}^3$.

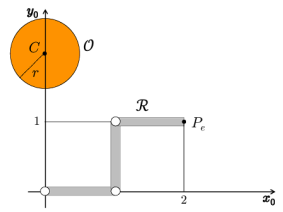

Figure 1: The 3R planar robot $\mathcal{R}$ with a single circular obstacle $\mathcal{O}$ in its workspace.

i) A desired Cartesian velocity $\boldsymbol{v}_e \in \mathbb{R}^2$ should be imposed to the end-effector point P_e of the robot $\mathcal{R}$, while maximizing the distance function $H_{\text{dist}}(\boldsymbol{q})$ between the robot body (i.e., its kinematic skeleton) and a circular obstacle $\mathcal{O}$ centered at $C = (0, 2)$ and of radius $r = 0.5$ m, as expressed by

$$H_{\text{dist}}(\boldsymbol{q}) = \min_{\boldsymbol{p}\in\mathcal{R},\boldsymbol{o}\in\mathcal{O}} \|\boldsymbol{p}(\boldsymbol{q}) - \boldsymbol{o}\| = \|P_m(\boldsymbol{q}) - O_m\|,$$

where $P_m(\boldsymbol{q})$ and O_m are the closest points, respectively on the robot and on the obstacle, when the robot is in the given configuration $\boldsymbol{q}$. Determine the symbolic expression and the numerical value of the velocity command $\dot{\boldsymbol{q}}_{PG}$ according to the Projected Gradient (PG) method when $\boldsymbol{v}_e = \begin{pmatrix} 0 & 1 \end{pmatrix}^T$ m/s. Verify your result and compute also the resulting velocity $\boldsymbol{v}_m \in \mathbb{R}^2$ of the point P_m.

ii) Suppose now that the robot $\mathcal{R}$ has two tasks assigned, ordered by priority: the first task is to impose the same previous velocity $\boldsymbol{v}_e$ to the point P_e; the second task is to impose the velocity $\boldsymbol{v}_m$ obtained in the previous item to the point P_m. Determine the symbolic expression and the numerical value of the velocity command $\dot{\boldsymbol{q}}_{TP}$ according to the Task Priority (TP) method. Do the joint velocities $\dot{\boldsymbol{q}}_{TP}$ and $\dot{\boldsymbol{q}}_{PG}$ have different directions in the joint space or not? Explain why. What if we choose $\boldsymbol{v}_m = \alpha\,(P_m - O_m)$, for some $\alpha > 0$, as desired velocity for P_m?

In the given configuration $q = (0, \frac{\pi}{2}, -\frac{\pi}{2})^T$, the closest point to the obstacle is the third joint, its position is:

$$P_m = \begin{cases} c_1 + c_{12} \\ s_1 + s_{12} \end{cases} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$ while the closest point of the obstacle is

$$O_m = C + \begin{pmatrix} r\cos\frac{\pi}{4} \\ -r\sin\frac{\pi}{4} \end{pmatrix} = \begin{pmatrix} \sqrt{2}/4 \\ 2 - \sqrt{2}/4 \end{pmatrix} \Rightarrow H_{dist}(q) = \left\| \begin{pmatrix} c_1 + c_{12} - \sqrt{2}/4 \\ s_1 + s_{12} - 2 + \sqrt{2}/4 \end{pmatrix} \right\|$$

the square is: $H_{dist}^2 = c_1^2 + c_2^2 + 2c_1c_2 - \sqrt{2}c_1 - \sqrt{2}c_2 + \frac{1}{4} + s_1^2 + s_2^2 + 2s_1s_2 - 2\left(2 + \frac{\sqrt{2}}{4}\right)s_1 - 2\left(2 + \frac{\sqrt{2}}{2}\right)s_2$

$$+ \frac{33 - 8\sqrt{2}}{8} = 2\cos(q_1 - q_2) - \sqrt{2}(c_1 + c_2) - \left(4 + \frac{\sqrt{2}}{2}\right)(s_1 + s_2) + 4.9607$$

Since minimizing the distance squared is equivalent to minimizing the distance, we use directly $H'(q) = H_{dist}^2(q)$.

$$\nabla H' = \left(-2\sin(q_1 - q_2) + \sqrt{2}s_1 - \left(4 + \frac{\sqrt{2}}{2}\right)c_1,\ 2\sin(q_1 - q_2) + \sqrt{2}s_2 - \left(4 + \frac{\sqrt{2}}{2}\right)c_2,\ 0 \right)^T$$

In the current conf, this evaluates in $\dot{q}_N = \nabla H'(q) = (-1.292,\ 3.414,\ 0)^T$

This is the preferred direction. The PG sol. is:

$$\dot{q}_{PG} = J^{\#} + (I - J^{\#}J)\dot{q}_N$$

where $J =$

$$P_m = \begin{cases} c_1 + c_2 \\ s_1 + s_2 \end{cases}$$

Exercise #2

Figure 2 shows a 2R robot moving under gravity in the 3D space, with the associated Denavit-Hartenberg (D-H) frames. The following assumptions are made on the position of the center of mass and on the barycentric inertia of the two links, when these quantities are expressed in the frame attached to the respective link[1]:

$$^1\boldsymbol{r}_{c1} = \begin{pmatrix} r_{c1,x} \\ r_{c1,y} \\ r_{c1,z} \end{pmatrix}, \quad ^2\boldsymbol{r}_{c2} = \begin{pmatrix} r_{c2,x} \\ 0 \\ 0 \end{pmatrix}, \quad ^1\boldsymbol{I}_{c1} = \begin{pmatrix} I_{c1,xx} & I_{c1,xy} & I_{c1,xz} \\ I_{c1,xy} & I_{c1,yy} & I_{c1,yz} \\ I_{c1,xz} & I_{c1,yz} & I_{c1,zz} \end{pmatrix}, \quad ^2\boldsymbol{I}_{c2} = \mathrm{diag}\{I_{c2,xx}, I_{c2,yy}, I_{c2,zz}\}.$$

Note in particular that the center of mass of the first link is not on the axis of joint 1.

i) Derive the dynamic model in the form

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau},$$

and a factorization matrix $\boldsymbol{S}$ such that $\boldsymbol{S}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} = \boldsymbol{c}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ and matrix $\dot{\boldsymbol{M}} - 2\boldsymbol{S}$ is skew-symmetric.

	α	a	d	θ
1	$-\pi/2$	0	l_1	q_1
2	0	l_2	0	q_2

The DH transformations are:

$$^0A_1 = \begin{pmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad ^1A_2 = \begin{pmatrix} c_2 & -s_2 & 0 & l_2 c_2 \\ s_2 & c_2 & 0 & l_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

i apply the recursive algorithm to compute T:

$i=1$)

$$^1\omega_1 = {}^0R_1^T\left(0 + (0 \quad 0 \quad \dot{q}_1)^T\right) = \begin{pmatrix} c_1 & s_1 & 0 \\ 0 & 0 & -1 \\ -s_1 & c_1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 0 \\ \dot{q}_1 \end{pmatrix} = \begin{bmatrix} 0 \\ -\dot{q}_1 \\ 0 \end{bmatrix}$$

$$^1v_1 = {}^0R_1^T\left(0 + (0 \quad 0 \quad \dot{q}_1)^T \times (0 \quad 0 \quad l_1)\right) = 0$$

$$v_{c1} = {}^1v_1 + {}^1\omega_1 \times r_{c1} = \begin{pmatrix} 0 & 0 & -\dot{q}_1 \\ 0 & 0 & 0 \\ \dot{q}_1 & 0 & 0 \end{pmatrix}\begin{pmatrix} r_{c1x} \\ r_{c1y} \\ r_{c1z} \end{pmatrix} = \begin{pmatrix} -\dot{q}_1 r_{c1z} \\ 0 \\ \dot{q}_1 r_{c1x} \end{pmatrix}$$

$$T_1 = \frac{1}{2} m_1 v_{c1}^T v_{c1} + \frac{1}{2} {}^1\omega_1^T I_{c1} {}^1\omega_1 = \frac{1}{2} m_1 \dot{q}_1^2 (r_{c1z}^2 + r_{c1x}^2) + \frac{1}{2}(0 \quad -\dot{q}_1 \quad 0)\begin{pmatrix} -\dot{q}_1 I_{c1xy} \\ -\dot{q}_1 I_{c1yy} \\ -\dot{q}_1 I_{c1yz} \end{pmatrix} =$$

$$\frac{1}{2} m_1 \dot{q}_1^2 (r_{c1z}^2 + r_{c1x}^2) + \frac{1}{2} I_{c1yy} \dot{q}_1^2 = \dot{q}_1^2 \left(\frac{1}{2}\left[m_1 (r_{c1z}^2 + r_{c1x}^2) + I_{c1yy}\right]\right)$$

$i=2$)

$$^2\omega_2 = {}^1R_2^T\left({}^1\omega_1 + (0 \quad 0 \quad \dot{q}_2)^T\right) = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ -\dot{q}_1 \\ \dot{q}_2 \end{pmatrix} = \begin{pmatrix} -\dot{q}_1 s_2 \\ -\dot{q}_1 c_2 \\ \dot{q}_2 \end{pmatrix}$$

$$^2v_2 = {}^1R_2^T\left({}^1v_1 + {}^1\omega_2 \times (l_2 c_2 \quad l_2 s_2 \quad 0)^T\right) = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -\dot{q}_2 & -\dot{q}_1 \\ \dot{q}_2 & 0 & 0 \\ \dot{q}_1 & 0 & 0 \end{pmatrix}\begin{pmatrix} l_2 c_2 \\ l_2 s_2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \dot{q}_2 l_2 \\ \dot{q}_1 l_2 c_2 \end{pmatrix}$$

$$v_{c2} = {}^2v_2 + {}^2\omega_2 \times r_{c2} = \begin{pmatrix} 0 \\ \dot{q}_2 l_2 \\ \dot{q}_1 l_2 c_2 \end{pmatrix} + \begin{pmatrix} 0 & -\dot{q}_2 & -\dot{q}_1 c_2 \\ \dot{q}_2 & 0 & \dot{q}_1 s_2 \\ \dot{q}_1 c_2 & -\dot{q}_1 s_2 & 0 \end{pmatrix}\begin{pmatrix} r_{c2x} \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \dot{q}_2 r_{c2x} + \dot{q}_2 l_2 \\ \dot{q}_1 r_{c2x} c_2 + \dot{q}_1 l_2 c_2 \end{pmatrix} = \begin{cases} 0 \\ \dot{q}_2 (r_{c2x} + l_2) \\ \dot{q}_1 c_2 (r_{c2x} + l_2) \end{cases}$$

$$T_2 = \frac{1}{2} m_2 v_{c2}^T v_{c2} + \frac{1}{2} {}^2\omega_2^T I_{c2} {}^2\omega_2 = \frac{1}{2} m_2 \left[\dot{q}_2^2 (r_{c2x} + l_2)^2 + \dot{q}_1^2 c_2^2 (r_{c2x} + l_2)^2\right] + \frac{1}{2}\left(\dot{q}_1^2 s_2^2 I_{c2xx} + \right.$$

$$\left. \dot{q}_1^2 c_2^2 I_{c2yy} + \dot{q}_2^2 I_{c2zz}\right) =$$

$$\dot{q}_1^2 \frac{1}{2}\left[m_2 c_2^2 (r_{c2x} + l_2)^2 + s_2^2 I_{c2xx} + c_2^2 I_{c2yy}\right] +$$

$$\dot{q}_2^2 \frac{1}{2}\left[m_2 (r_{c2x} + l_2)^2 + I_{c2zz}\right]$$

So at the end :

$$m_{11} = m_2 c_2^2 (r_{c2x} + l_2)^2 + s_2^2 I_{c2xx} + c_2^2 I_{c2yy} + m_1 (r_{c1z}^2 + r_{c1x}^2) + I_{c1yy} =$$

$$c_2^2 \underbrace{\left(m_2 (r_{c2x} + l_2)^2 + I_{c2yy} \right)}_{\rho_2} + \underbrace{s_2^2 I_{c2xx}}_{\rho_3} + \underbrace{m_1 (r_{c1z}^2 + r_{c1x}^2) + I_{c1yy}}_{\rho_1}$$

$$= \rho_2 c_2^2 + \rho_3 s_2^2 + \rho_1$$

$$m_{22} = m_2 (r_{c2x} + l_2)^2 + I_{c2zz} = \rho_4 \Rightarrow M = \begin{pmatrix} \rho_2 c_2^2 + \rho_3 s_2^2 + \rho_1 & 0 \\ 0 & \rho_4 \end{pmatrix}$$

Now i compute the Christoffel symbol : * $\frac{d}{dq_2}(c_2^2) = -2c_2 s_2$ $\quad \frac{d}{dq_2} s_2^2 = 2c_2 s_2$

$$c_{111} = \frac{1}{2}\left(\frac{\partial m_{11}}{\partial q_1} + \frac{\partial m_{11}}{\partial q_1} - \frac{\partial m_{11}}{\partial q_1} \right) = 0$$

$$\underset{c_{121}}{\overset{c_{112}}{\shortparallel}} = \frac{1}{2}\left(\frac{\partial m_{11}}{\partial q_2} + \frac{\partial m_{12}}{\partial q_1} - \frac{\partial m_{21}}{\partial q_1} \right) = 2c_2 s_2 (\rho_3 - \rho_2)$$

$$c_{122} = \frac{1}{2}\left(\frac{\partial m_{12}}{\partial q_2} + \frac{\partial m_{12}}{\partial q_2} - \frac{\partial m_{2}}{\partial q_1} \right) = 0$$

$$c_{211} = \frac{1}{2}\left(\frac{\partial m_{21}}{\partial q_1} + \frac{\partial m_{21}}{\partial q_1} - \frac{\partial m_{11}}{\partial q_2} \right) = 2c_2 s_2 (\rho_2 - \rho_3)$$

$$\underset{c_{212}}{\overset{c_{221}}{\shortparallel}} = \frac{1}{2}\left(\frac{\partial m_{22}}{\partial q_1} + \frac{\partial m_{21}}{\partial q_2} - \frac{\partial m_{12}}{\partial q_2} \right) = 0$$

$$c_{222} = \frac{1}{2}\left(\frac{\partial m_{22}}{\partial q_2} + \frac{\partial m_{22}}{\partial q_2} - \frac{\partial m_{22}}{\partial q_2} \right) = 0$$

$\Rightarrow$ I compute the C matrix

$$C_{11} = c_{111} + c_{121} = 2c_2 s_2 (\rho_3 - \rho_2) \qquad C_{12} = c_{112} + c_{122} = 2c_2 s_2 (\rho_3 - \rho_2)$$

$$C_{21} = c_{211} + c_{221} = 2c_2 s_2 (\rho_2 - \rho_3) \qquad C_{22} = c_{212} + c_{222} = 0$$

$$\Rightarrow C(q, \dot{q}) = \begin{pmatrix} 2c_2 s_2 (\rho_3 - \rho_2) & 2c_2 s_2 (\rho_3 - \rho_2) \\ 2c_2 s_2 (\rho_2 - \rho_3) & 0 \end{pmatrix} \begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \end{pmatrix} = \begin{cases} 2c_2 s_2 (\rho_3 - \rho_2)(\dot{q}_1 + \dot{q}_2) \\ 2c_2 s_2 (\rho_2 - \rho_3) \dot{q}_1 \end{cases}$$

For the gravity term, i need the position of the centers of mass.

$$^0 r_{c1} = {}^0A_1 \, {}^1 r_{c1} = \begin{pmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} r_{c1x} \\ r_{c1y} \\ r_{c1z} \\ 1 \end{pmatrix} = \begin{pmatrix} r_{c1x} c_1 - r_{c1z} s_1 \\ r_{c1x} s_1 + r_{c1z} c_1 \\ -r_{c1y} + l_1 \end{pmatrix}$$

$$^0 r_{c2} = {}^0A_1 \, {}^1A_2 \, {}^2 r_{c2} = \begin{pmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_2 & -s_2 & 0 & l_2 c_2 \\ s_2 & c_2 & 0 & l_2 s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ r_{c2x} \\ 1 \end{pmatrix} =$$

$$\begin{pmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} l_2 c_2 \\ l_2 s_2 \\ r_{c2x} \\ 1 \end{pmatrix} = \begin{pmatrix} l_2 c_1 c_2 - r_{c2x} s_1 \\ l_2 s_1 c_2 + r_{c2x} c_1 \\ l_1 r_{c2x} - l_2 s_2 \end{pmatrix}$$

The gravity vector in RF_0 is $(-g_0 \quad 0 \quad 0)^T$

$$U = -m_1 {}^0g^T {}^0r_{c1} - m_2 {}^0g^T {}^0r_{c2} = m_1 g_0 (r_{c1x} c_1 - r_{c1z} s_1) + m_2 g_0 (l_2 c_1 c_2 - r_{c2x} s_1)$$

the gravity term is

$$\nabla_q U = \begin{bmatrix} m_1 g_0 (r_{c1z} c_1 - r_{c1x} s_1) - m_2 g_0 (l_2 s_1 c_2 + r_{c2x} c_1) \\ - m_2 g_0 l_2 c_1 s_2 \end{bmatrix}$$

Model:

$$\begin{pmatrix} \rho_2 c_2^2 + \rho_3 s_2^2 + \rho_1 & 0 \\ 0 & \rho_4 \end{pmatrix} \ddot{q} + \begin{pmatrix} 2 c_2 s_2 (\rho_3 - \rho_2)(\dot{q}_1 + \dot{q}_2) \\ 2 c_2 s_2 (\rho_2 - \rho_3) \dot{q}_1 \end{pmatrix} +$$

$$\begin{pmatrix} m_1 g_0 (r_{c1z} c_1 - r_{c1x} s_1) - m_2 g_0 (l_2 s_1 c_2 + r_{c2x} c_1) \\ - m_2 g_0 l_2 c_1 s_2 \end{pmatrix} = \tau$$

Exercise #4

Consider a rigid robot manipulator with n revolute joints, without external contact forces/moments and dissipative effects. The components of the motor torque $\boldsymbol{\tau} \in \mathbb{R}^n$ are bounded as

$$|\tau_i| \leq T_i, \qquad \text{with } T_i \geq 5 \cdot \max_{\boldsymbol{q}} |g_i(\boldsymbol{q})|, \qquad i = 1, \ldots, n,$$

namely with bounds that are large enough to guarantee that the robot always sustains at least its own weight under gravity (here, with a conservative margin factor of 5). Let the robot state at time $t = t_0$ be $\boldsymbol{x}(t_0) = (\boldsymbol{q}(t_0), \dot{\boldsymbol{q}}(t_0))$, with a total robot energy $E(t_0) = E_0$.

i) If the robot is in a generic state $\boldsymbol{x}(t_0) = (\boldsymbol{q}_0, \dot{\boldsymbol{q}}_0)$ with $\dot{\boldsymbol{q}}_0 \neq \boldsymbol{0}$, choose the torque $\boldsymbol{\tau}_0 = \boldsymbol{\tau}(t_0)$ to be delivered by the motors so that the total energy E will instantaneously decrease as much as possible.

ii) Suppose now that the robot is in a state $\boldsymbol{x}(t_0) = (\boldsymbol{q}_0, \boldsymbol{0})$, namely **at rest** and in a configuration $\boldsymbol{q}_0$ such that $\boldsymbol{g}(\boldsymbol{q}_0) \neq \boldsymbol{0}$. Choose again the motor torque $\boldsymbol{\tau}_0$ so that the total energy E will decrease. *Hint:* If this is not possible, choose $\boldsymbol{\tau}_0$ so that $\dot{E}$ will instantaneously decrease.

The model without any dissipative effects is just:

$M(q)\ddot{q} + c(q, \dot{q}) + g(q) = \tau$. We know that

$\dot{E} = \dot{q}^T \tau \Rightarrow \dot{q}^T(0)\tau$ to find the torque that leads to the biggest decrease, we need to find the τ that minimizes $\dot{q}_0^T \tau$ over the set $|\tau_i| < T_i$. This is a linear program:

$$\begin{cases} \min\limits_{\tau} & q_0^T \tau \\ \text{s.t.} & |\tau_i| < T_i \quad \forall i \end{cases}$$

so the solution can be found with the classical algorithms like the simplex method.

If $\dot{q}_0 = 0$ then $\forall \tau : \dot{E} = \dot{q}_0^T \tau = 0$ so it is impossible at the given state to instantaneously decrease the energy $E = T + U$.

We can try to decrease $\dot{E}$ by considering $\ddot{E} = \ddot{q}^T \tau + \dot{q}^T \dot{\tau} = \ddot{q}^T \tau$

At zero velocity the model is

$$M\ddot{q} + g(q) = \tau \Rightarrow \ddot{q} = M^{-1}(\tau - g) \Rightarrow \ddot{E} = \left[M^{-1}(\tau - g)\right]^T \tau = (\tau - g)^T M^{-T} \tau$$

So we have to solve:

$$\begin{cases} \min\limits_{\tau} & (\tau - g)^T M^{-T} \tau = \tau^T M^{-T} \tau - g^T M^{-T} \tau \\ \text{s.t.} & |\tau_i| < T_i \quad \forall i \end{cases}$$

is a quadratic form.